

Global Synthetic Labor Index Prototype

Identity → Proof → Settlement → Markets

A civilization-grade control loop for verifiable synthetic work:
Observe → Prove → Settle → Print → Price → Finance → Expand



One invariant. Four pillars.

No value without evidence · No settlement without validation

Identity

Roles bound to keys (ENS/DID).
Accountability by default.
Scope rules prevent ambiguity.

Evidence

Deterministic runs emit proof bundles.
Replay beats rhetoric (audit = replay).
Commit-reveal reduces bribery leverage.

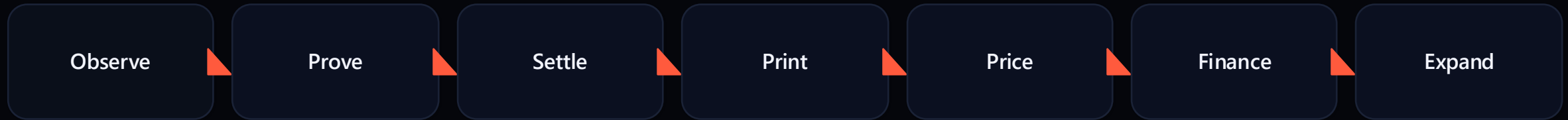
Settlement

Escrow releases only after validation.
Receipts are settlement-grade artifacts.
No payout without PoC.

Governance

Policy gates, upgrades, emergency brakes.
Owner override is final (bounded autonomy).
MethodHash freezes index rules per epoch/quarter.

A low-entropy path where honesty wins.



Physics intuition (Gibbs free energy)

$$F = U - T \cdot S$$

U: utility captured · S: ambiguity/dispute surface

T: operational friction

Goal: drive $T \cdot S \rightarrow 0$ via determinism + receipts + incentives.

Game theory (dominant strategies)

Workers earn only for validated proof.

Validators are stake-bonded; misbehavior becomes negative-sum (slashing).

Commit-reveal reduces herding and bribery leverage.

Ledger & Governance

Freeze truth at L0. Evolve policy at L1—without rewriting history.



L0 Aegis · L1 Apotheosis

Conservation laws + surgical brakes

L0 Aegis (frozen)

Commit · Receipt · Settle — reconstructible from events.

Invariants:

- Conservation of value
- Single finality (idempotent)
- Deterministic dispute windows
- Receipt anti-replay (EIP-712 + nonces)
- Role legibility (ENS ↔ key)

L1 Apotheosis (evolvable)

Policy modules are hot-swappable but may only restrict behavior—never violate L0.

Pause lattice:

- JOBS: stop new commitments
- RECEIPTS: stop new submissions
- PAYOUTS: stop fund egress
- DISPUTES: freeze adjudication

Identity & Safety

One signature, bounded blast radius: fuse membranes + witnessed delegation + recovery.



ENS namespace grammar

One rule + scoped meaning (low entropy, high legibility)

```
<entity>.(<env>.)<role>.agi.eth  
role ∈ {club, agent, node} · env ∈ ENV_SET (optional)
```

Role lattice (alpha + production)

Validators: *.alpha.club.agi.eth | *.club.agi.eth
Agents: *.alpha.agent.agi.eth | *.agent.agi.eth
Nodes: *.alpha.node.agi.eth | *.node.agi.eth
Sovereigns: *.alpha.agi.eth | *.agi.eth

Guardrail: reserve {agent,node,club} under every env.agi.eth.

Scale without ambiguity

ENSIP-10 wildcards for *.role namespaces.
EIP-3668 (CCIP-Read) for high-churn metadata.
ENSIP-15 normalization to harden lookups.

Best practice: keep trust anchors minimal; push churn into resolvers.

AGI Passport (EIP-7702 safety layer)

Identity membrane · delegation transparency · auto-healing recovery

Identity Membrane

DTL / DTLL

Quarantine

Recover

Re-onboard

Delegation Transparency Fabric

DTL-1: append-only Merkle log of delegation events.
DTLL-1: sovereign-signed log list (wallet-consumable).
Wallet rule: ≥ 2 independent logs + $\geq t$ witnesses, or no delegation UX.

Fuse membranes + recovery

Fuse presets (role): CANNOT_UNWRAP · CANNOT_TRANSFER · CANNOT_SET_RESOLVER · ...
Rescue Seed Vault: pre-signed kill-switch (clear delegation).
SRD: sterile quarantine delegation (no outbound execution).

Routing invariant: No Verified Delegation → No Sponsored Gas → No Job Routing.

Evidence & Observatory

Proof-of-Contribution turns telemetry into economic evidence; the index becomes auditable truth.



Proof-of-Observability → Proof-of-Contribution

Observe → commit → attest → challenge → settle → index



Inclusion predicate (mechanical)

A job enters the macrostate only if:

- (1) escrow settled,
- (2) valid validator VSA signature(s),
- (3) survives challenge window.

No PoC → no inclusion (and eventually: no payout).

```
struct VSA {  
    address jobManager;  
    uint256 jobId;  
    address worker;  
    address validator;  
    bytes32 jobSpecHash;  
    bytes32 inputHash;  
    bytes32 outputHash;  
    bytes32 obsRoot;  
    bytes32 artifactsRoot;  
    uint256 alphaWU;  
    uint16 qualityScore;  
    uint64 deadline;  
    uint256 nonce;  
}
```

α -Work Units (α -WU)

A canonical, auditable unit of verified synthetic labor

$$\alpha\text{-WU} = \text{VerifierGas}^\theta \times \text{Duration}^{(1-\theta)} \times \text{QualityScore} \quad (\theta \approx 0.5 \text{ v0})$$

Metrology rules (v0)

Count only PoC-attested settlements.
Fails acceptance/SLO $\rightarrow$ 0 credit.
Robust stats by default (median, winsorize).
Normalize using rolling windows (e.g., 13 weeks).

Token coupling (thermostat)

Signals: validated α -WU, disputes, SLO drift, burn/emission ratios.
Actuators: fee splits, burn fraction, tier multipliers, quorum/slashing.
Goal: stability = throughput with bounded risk.

Global Synthetic Labor Index (GSLI) v0

Weekly macro observable: output × price × reliability × breadth

Constituents

Q: $\sum \alpha$ -WU (PoC settlements)
P: median AGIALPHA per α -WU
R: uptime + SLA – disputes
B: entropy over agents/nodes/sectors

$$\text{GSLI}_t = 100 \cdot \exp(0.55 \cdot \Delta \ln Q + 0.25 \cdot \Delta \ln P + 0.15 \cdot \Delta \ln R + 0.05 \cdot \Delta \ln(1+B))$$

Index prints (on-chain anchor)

Public truth with private compute. Store only: weekId, datasetRoot (Merkle), methodHash, indexValue, publisherSig, challengeWindow.
Any auditor can recompute from datasetRoot. Weights locked per quarter; changes bump methodHash.

Synthetic Labor Observatory

A public, reproducible oracle for civilization-scale work

Metrology

Measure α -WU as events.
Outputs: acceptance, quality,
on-time, yield $\rightarrow$ GSLI.

Integrity Reserve

Every WU pays audit premium.
Public RNG selects deep audits.
Proper scoring rules pay
auditors.

Truth Modules

Deterministic replay.
Evaluator harness.
Proof-carrying work
(zkVM/SNARK).
Truth > consensus.

Proof-First

Proofs settle cheaply; audits
become safety net.
Prover Market commoditizes
truth throughput.

Markets & Macro

Once work is measured and provable, it becomes financeable: curve → repo → corridor → credit.



Synthetic Labor Yield Curve

Term structure of verified output (α -WU credits)

Instrument: LZ(T) labor zeros (v0)

1 LZ(T) redeems for 1 verified α -WU credit during delivery window at maturity T.
Price today: $Z(T)$ (AGIALPHA per credit).
Implied yield: $rL(T) = (1/T) \cdot \ln(S_0/Z(T))$.

Launch tenors (Tier A first)

7D · 14D · 30D · 90D · 180D · 365D
Spot index $S(t)$ uses PoC-only settlements.
Reliability becomes a priced spread via tiering.

Illustrative curve (not market data)



Labor-Backed Compute Notes

Finance expansion by selling long-dated verified output

Bond + capacity

Mint LZ(180/365)

Auction

Capex (GPUs)

Deliver PoC α -WU

Redeem

Risk controls (default dominated)

Issuance limit: quantile throughput $\times \beta$ safety factor.
Performance bond + penalty buy-in on shortfall.
Only external demand counts toward delivery (anti-wash).
Tiered eligibility based on reliability history.

Macro flywheel

Term structure $\rightarrow$ financing $\rightarrow$ more capacity $\rightarrow$ lower volatility $\rightarrow$ deeper curve $\rightarrow$ cheaper capital.
PoC + slashing + buy-in enforce delivery.

Labor Repo Market

Make long-dated credits liquid collateral; create the short end

Two venues (launch order)

- 1) Pool facility: always-on borrow/lend vs LZ_A_T collateral.
- 2) Term repo auctions: O/N, 7D, 30D benchmark fixings.
Repo rates become the short end of the curve.

Conservative oracle + haircuts

$P_{\text{oracle}} = \min(\text{median}(\text{auction}, \text{TWAP}, \text{curve}), \text{auction})$.
Haircuts widen with volatility, illiquidity, concentration.
Auto-convert near maturity into spot credits to avoid expiry risk.

Suggested starting parameters (Tier A)

Collateral	cf	lt	liqBonus
LZ_A_365	0.55	0.62	6%
LZ_A_180	0.62	0.70	5%

cf = collateral factor · lt = liquidation threshold

Standing Facility (Labor Open-Market Ops)

Corridor policy: cap stress, absorb excess liquidity

Two standing windows

Ceiling: lend AGIALPHA vs SR/SRCP at penalty rate.
Floor: take AGIALPHA via escrow-collateralized reverse repo.
Optional: bounded OMO buys/sells to shape the long end.

Rule-bound policy (best practice)

Set r^* from observables: labor inflation π_L , output growth g_Q , stress σ .
Corridor widens as σ rises (avoid dominance).
Hard caps; emergency actions only widen haircuts / pause new credit.

Short rates anchor everything: repo $\rightarrow$ paymaster APR $\rightarrow$ curve stability

FLOOR

MARKET

CEILING



Universal Job Paymaster → Receivables

One signature → job funded → PoC-gated payout → repay later



Underwriting (v0)

Start overcollateralized (AGIALPHA and/or LZ_A_T).
Limits derived from conservative haircuts.
Scores improve only from external-demand jobs (anti-wash).
No PoC → no payout → no receivable.

Pricing model

$r_{\text{Borrow}} = r_{\text{Base}}(\tau) + s_{\text{Risk}}(\text{borrower}) + s_{\text{Job}}(\text{class})$
Base rate anchored by corridor (floor/ceiling).
Delinquency triggers freeze + liquidation.
Receivables can be pooled into SR/JR tranches.

Invoices → Money-Market Instruments

Pool PoC-final receivables · mint SR/JR · finance SR in repo

Pooling constraints (money-market posture)

Only PoC-final invoices (settled; attestation survived challenge).
WAM $\leq$ 30d, WAL $\leq$ 60d; borrower/cluster caps.
Reserve account + junior retention (first loss).
Coverage tests (OC/IC/DQ) divert cash to SR when stressed.

Tranches

SR (Senior): targets short-duration “cash-like” behavior.
JR (First loss): absorbs losses; originator retains 5–15%+.
Wrap SR into SRCP_7D / SRCP_30D for clean repo collateral.

Policy collateral

Use SR/SRCP as the standing facility’s core collateral: lend at the ceiling, absorb at the floor.
Result: coherent macro regime where short rates, credit, and labor prices reference the same verifiable state variables.

Epoch Snapshot Market

From procurement auctions → continuous clearing → ERC-1155 coverage futures



Treat liveness as inventory, not bidding luck.

Predictable coverage over a rolling horizon; backups make liveness a guarantee.

Failure modes of discrete procurement

Volatility (thin bids → spiky prices) · Liveness cliffs (no winner or winner fails) · Wasted compute (many prove; one paid) · Paymaster stress (retries amplify sponsored gas burn).

Epoch auction

Continuous offers

Coverage futures (1155)

Capacity vaults (4626)

Design target

Constant-cost verification on-chain; heavy work off-chain.
Best-K offers stored; deterministic selection.
Rolling buffer + rescue paths ensure “never stall”.

Gibbs framing

Drive $T \cdot S \rightarrow 0$: reduce friction × ambiguity via determinism + replayable receipts + incentives.

Epoch snapshots: the admission macrostate

Join intents → batch proof → epoch root → cheap membership checks

JoinBook

Batch proof (zkVM)

EpochAdmission root

Merkle gate + VSA

ERC-1155 coverage slots (futures)

Slot = tradable obligation to deliver (root + proof + metaDigest).

tokenId = keccak256(chainId, epochId, lane).

Lanes: PRIMARY (must deliver) + BACKUP (paid from slashed margin) + optional BACKUP2.

Anti-theft: proof binds to slotOwner (public input).

Paymaster rules (safe sponsorship)

Sponsor only deterministic, non-reverting paths.

Allowlist selector: TournamentHub.attest(...).

Staticcall verifyLeaf(epochId, leaf, proof) before sponsoring.

Bound calldata size; per-user + global budgets; epoch rate limits.

Lineage & Policy

Sovereign Genesis Charter · registry-as-genome · shadow-fork policy loop



First lineage asset: Sovereign Genesis Charter

ENS-anchored · multisig-controlled · manifest-pinned

Why this wins (Schelling point)

Collapses ambiguity: everyone agrees where "truth" lives.
Upgrades remain legible; forks retain provenance.
Role issuance is deterministic; recovery is predefined.
Transfer is a ceremony (timelocked + audited).

```
{  
  "epoch": "alpha",  
  "root_ens": "genesis.alpha.agi.eth",  
  "owner_multisig": "0x...",  
  "role_namespaces": {  
    "validators": "*.alpha.club.agi.eth",  
    "agents": "*.alpha.agent.agi.eth",  
    "nodes": "*.alpha.node.agi.eth"  
  },  
  "policy_hash": "keccak256(...)"  
}
```

Shadow-fork policy loop (owner-gated execution)

Alert opens → fork mainnet @ block → simulate tx bundle → score deltas → propose → owner-signed execute only.
Guardrails: liveness $\geq$ floor; receipts required; non-reversible actions only via timelock.

Verifiable Anchors

Contracts · ENS namespaces · onboarding rails · standards



Publicly verifiable anchors

Verify domain + contract + decimals before signing

Ethereum contracts (Etherscan)

AGIJobManager

0x0178B6baD606aaF908f72135B8eC32Fc1D5bA477

AGIALPHA (transitory, 6 decimals)

0x2e8fb54c3ec41f55f06c1f082c081a609eaa4ebe

AGIALPHAEqualMinterVault

0x27d6fE8668c6f652Ac26fFaE020D949f03aF80D8

AGIALPHA v2 (final, 18 decimals)

0xA61a3B3a130a9c20768EEBF97E21515A6046a1fA

Key links

- [AGIJobManager \(Etherscan\)](#)
- [AGIALPHA v2 \(Etherscan\)](#)
- [EIP-712](#)
- [EIP-3668 \(CCIP-Read\)](#)
- [ENSIP-10 \(wildcards\)](#)
- [deBridge \(SOL ↔ ETH rails\)](#)
- [Chainlink VRF \(audit lottery RNG\)](#)

ENS roles

Validators: *.alpha.club.agi.eth | *.club.agi.eth
Agents: *.alpha.agent.agi.eth | *.agent.agi.eth
Nodes: *.alpha.node.agi.eth | *.node.agi.eth
Sovereigns: *.alpha.agi.eth | *.agi.eth

Build the Cathedral of Verifiable Work

Autonomy measured · Work proven · Value settled

Simple path:

1) Prove work (PoC) 2) Print truth (GSLI) 3) Price time (LZ curve) 4) Finance scale (repo + corridor) 5) Guarantee liveness (epoch coverage)

AGI Passport — EIP-7702 Safety Layer (source)

Verbatim source slides (compressed).

AGI Passport

One-Signature Onboarding • EIP-7702 Safety Layer • ENS Fuse Membranes

"As Columbus braved the seas, we now venture into the vast oceans of AGI.
The age of AGI exploration has dawned."

— to AGI Agent

AGI Passport • v1

agi.eth

#1

The System in Three Pillars

Reduce onboarding friction without reducing safety: identity, transparency, recovery.



Identity Membrane

ENS names + fuse presets create an unforgeable role identity: what can change, what cannot.



Delegation Transparency

Every EIP-7702 delegation becomes an auditable event: Merkle logs + witnessed STHs.



Auto-Healing Recovery

Pre-signed kill-switch + quarantine delegation: recovery works even if the user is offline.

Design goal: one approval, bounded state radius

Practice • EIP-7702 • DTL

#2

Role Lattice (ENS Namespaces)

Roles are human-readable, machine-enforceable, and fuse-hardened.

Agents

* alpha.agent.agi.eth
* agent.agi.eth

Execute tasks, hold preferences, sign outputs. Least privilege by default.

Nodes

* alpha.node.agi.eth
* node.agi.eth

Run jobs, post proofs, earn rewards. Bonded, monitored, slashable.

Validators

* alpha.club.agi.eth
* club.agi.eth

Witness STHs, attest safety, approve rescues. Protocol immune system.

Sovereigns

* alpha.agi.eth
* agi.eth

Sign DTL, log lists and global policy results. Root-of-trust keys.

Names connect roles, fuses enforce it.

ENS

#3

AGI Passport Kernel

An identity cell: authority + name + policy + attestations + commitments.

PASSPORT

Identity membrane
+ policy kernel

ENS Name

Human readable identity with resolver + records, fuse-locked.

Authority

EIP-7702 / AA smart-account behavior, session keys, guardians.

Policy

Allowed call graph, spend limits, role constraints, upgrade rules.

Attestations

Uptime, version, audits, jurisdiction flags, reputation proofs.

Commitments

Bond / stake, slashing conditions, unbonding windows.

Passport = stable identity + bounded execution

Kernel

#4

One-Signature Onboarding (L1 → L2)

A single approval expands capability while keeping policy tight.

Wallet

User approves
(EIP-7702 auth)

Verified Delegator

Safe code
allowed/revoked

Batch

ENS name
+ fuses
+ role

Commit

Bond/stake
+ keys
+ attest

Mirror

L2 registry
activation

Outcome: the account becomes a programmable passport with a constrained action space (fuses + policy), plus an auditable delegation record (DTL) and a recovery path (Seed Vault + SRD).

One approval. Many steps. Some ambiguity.

Onboarding

#5

ENS Fuse Membrane

Irreversible constraints that prevent identity rug-pulls and resolver tampering.

Representative fuses (preset by role)

CANNOT_UNWRAP

Effect

Locks the wrapped name; prevents bypass of wrapper restrictions.

CANNOT_TRANSFER

Makes the name non-transferable (identity cannot be sold or stolen by transfer).

CANNOT_SET_RESOLVER

Freezes resolver to prevent stealth redirection of records.

CANNOT_CREATE_SUBDOMAIN

Stops uncontrolled issuance of sub-identities.

Best practice: lock the high-risk surfaces permanently, and treat expiry as a renewal hygiene obligation.

Fuses are one-way valves: set them like you mean it.

ENS Wrapper

#6

AGI Passport — EIP-7702 Safety Layer (source)

Verbatim source slides (compressed).

Why EIP-7702 Needs an “Immune System”

Delegation is powerful: the blast radius must be bounded by policy, logs, and recovery.

Risks (if ungoverned)

- Pushing a single authorization tuple installs malicious execution.
- Delegation state can persist even if a transaction reverts.
- Upgradable delegates can “time-bomb” an identity.
- Chain-agnostic authorizations expand replay surface.

Controls (this deck)

- Allowlisted, audited delegates only (codehash pinning).
- DTL transparency logs + witnessed STHs.
- DTLL signed log lists: ≥2 independent logs required.
- Rescue Seed Vault + Sterile Recovery Delegator.

Wallet rule: If the safety checks fail, do not show a “normal” authorization flow.

Bounded blast radius in the product.

Threat Model

#7

7702 Safety Layer (Protocol View)

A firewall: registry + paymaster gating + quarantine + proofs.

Safety Registry

Allowlist delegates
Pin codehash
Role-specific policy

Wallet / Paymaster Gate

Require DTL
Require ≥2 logs
Require ≥1 witnesses

Quarantine + Recovery

Seed Vault kill-switch
Sterile delegation
Re-onboard safely

Enforced invariant:

No Verified Delegation → No Sponsored Gas → No Job Routing → No L2 Activation

This makes unsafe states economically cold; they cannot metabolize network resources.

Security is a routing decision, not a suggestion.

Safety Layer

#8

DTL-1: Delegation Transparency Log

A Merkle log for EIP-7702 delegation events: observable, provable, monitorable.

Append-only Merkle log

Leaves: [authority, delegate, nonce, txhash, codehash, ...]
Proof: [H1] H12 Head [DTH] H24

L1 L2 L3 L4

STH signed by log + ≥1 witnesses

Wallet enforcement rule

- If the delegation is not visible in the transparency fabric, treat it as unsafe.
- Require inclusion proofs to the latest witnessed STH.
- Alert on unexpected delegates / codehash drift.
- Quarantine on anomalies; recover via kill switch.

DTL makes “delegation state” a shared reality.

Transparency is the arbiter to alert delegation.

DTL-1

#9

DTLL-1: Signed Log List (Wallet-Consumable)

Sovereigns sign which logs exist; validators witness STHs; wallets require ≥2 independent logs.



Sovereigns sign DTL

Defines approved logs + operators + witness sets + policy defaults.



Two independent logs (operator diversity)

Wallet selects ≥2 logs with distinct operators; verifies STH signatures + consistency.



≥1 validator witnesses

Wallet requires ≥1 witness signatures on each log's STH before allowing a 7702 authorization UI path.

No witnessed transparency → no delegation UI

DTLL-1

#10

Rescue Seed Vault (Pre-Signed Kill Switch)

Recovery that works even if the user is offline during an incident.



What is stored

- Encrypted bundle of pre-signed EIP-7702 authorizations
- Only allowed action: delegate → Gd (clear code)
- Optional: delegate → SRD (sterile quarantine)
- Windowed by nonce range + expiry

Invariant: never store anything that can move funds.



How it is released

- Incident detected (watchtower / user / guardian)
- Validators approve rescue threshold
- Key shares revealed → decrypt bundle
- Relay broadcasts “clear delegation”
- Account returns to clean state

Anomalous detected → quarantine → heal → report

Rescue

#11

Sterile Recovery Delegator (SRD)

A quarantine delegation target: accept inbound assets, prevent outbound execution.



Allows

- Receive ETH
- Receive ERC-721 / ERC-1155
- ERC-1271 signature verification
- Explicit “re-onboard” ceremony



Forbids

- No arbitrary execution
- No token approvals
- No upgrades / upgrades
- No outbound value movement

Purpose:

Freeze the account into a safe, minimal state while the network executes recovery and key rotation.

Quarantine is a feature: it buys time without bleeding value.

SRD

#12

AGI Passport — EIP-7702 Safety Layer (source)

Verbatim source slides (compressed).

Operational Loop: Prevent → Detect → Recover

Security becomes a closed-loop control system.



Result: honest behavior becomes the easiest path; attacks become visible, expensive, and reversible.

Security is a control loop, not a checklist.

Ops

#13

Alpha → Production Hardening

Use alpha namespaces for agility, production namespaces for permanence.

Alpha namespaces

```
* alpha-agent.agi.eth  
* alpha-node.agi.eth  
* alpha-club.agi.eth  
* alpha.agi.eth
```

- Faster iteration
- Controlled revocation
- Conservative irreversible fuses
- Aggressive telemetry & canaries

Production namespaces

```
* agent.agi.eth  
* node.agi.eth  
* club.agi.eth  
* agi.eth
```

- Strong emancipation
- Permanent fuse membranes
- Strict DTL enforcement
- Formal rescue governance

Move to production when drawbacks are proven.

Phase

#14

Next Move: Make Wallets Refuse Unsafe Delegation

Adopt DTL gating as a public standard: witnessed transparency or nothing.

Ship this as a public good:

- DTL-1: A Merkle transparency log format for delegation events.
- DTL-1: A signed log list format (like CT log lists).
- Witness API: validator co-signing of STHs (threshold by role).
- Rescue protocol: Seed Vault + SRD as a standard quarantine path.

When a user clicks "Authorize", the wallet must already have:
#2 independent logs + #1 witnesses + verified delegate codehash.

If it can take your keys, it must be transparently witnessed.

Standardization

#15

AGI Passport

Identity that scales to a civilization.

One signature.
Unruggable ENS identity.
Witnessed delegation transparency.
Auto-healing recovery.

Deploy: Agents • Nodes • Validators • Sovereigns

Open to explore

in AGI

#16

Synthetic Labor Observatory (source)

Verbatim source slides (compressed).

Synthetic Labor Observatory

A public, reproducible oracle for civilization-scale synthetic work.
Identity → Proof → Settlement → Governance

GSLI + α-WU Metrology

Integrity Reserve + Audit Lottery

Truth Modules

Proof-First + Prover Market

For all sentient beings • Kantashov-II compatible measurement & coordination

#1

One invariant, four pillars

A trust engine that converges toward reality

Metrology
 Measure every work unit (α-WU) as verifiable events.
 Outputs: acceptance, quality, timeliness, yield → GSLI.

Integrity Reserve
 Every WU pays an audit premium.
 Random deep audits benchmark validators against truth.

Truth Modules
 Physics-tight verification: replay, evaluators, and proof-carrying work.
 Truth > social agreement.

Proof-First
 Proofs settle work cheaply and fast.
 Prover Market commoditizes truth throughput.

Invariant: reported metrics must be reproducible from public logs and anchored to audited reality (Q(d)).

#2

ENS role lattice

Role-gated telemetry + legible accountability

Validators
 *alpha.club.agi.eth / *.club.agi.eth
 Stake + commit/reveal + slashing

Agents
 *alpha.agent.agi.eth / *.agent.agi.eth
 Execute WUs; optionally submit proofs

Nodes
 *alpha.node.agi.eth / *.node.agi.eth
 Orchestrate WUs; emit telemetry

Resolution + offchain scale
 ENSIP-10 wildcard resolution enables namespaces like *.club.agi.eth.
 EIP-3668 (CCIP-Read) enables secure offchain lookups to scale subname metadata and telemetry without losing verifiability.
 ENSIP-15 normalization hardens name handling (anti-spoofing).
 resolveip(s) -> identity.keys.endpoints.com in \$!

#3

End-to-end work lifecycle

From opportunity → proof → settlement

Post
Escrow fee + audit premium

Assign
Node selects agent

Execute
Agent performs WU

Validate
Stake-weighted committee

Audit
Lottery deep checks

Settle
Payout + reputation

Telemetry → Metrology → Index

Every stage emits public events (who, what, when, stake, score). The Observatory aggregates these into GSLI and publishes a reproducible daily report.

Audit truth feeds back into Q(d) and validator incentives (β), continuously benchmarking "votes" against reality.

#4

α-WU Metrology → GSLI

The observable becomes the oracle

Four reproducible KPIs (from event logs)
 1) Acceptance rate
 2) Stake-weighted quality
 3) On-time completion
 4) Slashing-adjusted yield
 All computable from public telemetry; no hidden inputs.

GSLI (Global Synthetic Labor Index)
 A composite, versioned index over the KPI vector + volume + diversity.
 Published daily as a signed report + on-chain anchor (CID/hash).
 ReportGSLI = PF5 CD + on-chain pointer
 GSLI(s) = [accept, quality, onTime, yield, volume, diversity]

#5

Integrity Reserve + Audit Lottery

Continuous benchmarking by reality

Audit premium (every WU)
 Every work unit pays a tiny premium into the reserve.
 Reserve funds deep verification, offchain resolution, and insurance.
 WU fee + audit premium -> Integrity Reserve
 Lottery selection + deep verification
 Proper scoring rules pay auditors
 Q(d) anchored to audit truth

Lottery selection (public RNG)
 A verifiable RNG selects WUs for deep audit.
 Auditors are paid via proper scoring rules against revealed truth.
 Q(s) anchored to audit truth (postconsensus)

#6

Synthetic Labor Observatory (source)

Verbatim source slides (compressed).

VERIFICATION

Truth Modules

From audited opinions → audited facts

Deterministic replay
 Immutable manifest + container digest
 + canonical output hash.

 Auditors replay inside sandbox (VM/container) and submit signed receipts.

Evaluator harness
 Unit tests, benchmark harnesses, schema checks.

 Deterministic score() produces pass/fail + continuous score.

Proof-carrying work
 Agent submits succinct proof (ZKVM/SHARK).

 On-chain verifier checks proof binds to manifestHash + outputCanonicalHash.

Design rule: on-chain verifies proofs/receipts; execution happens off-chain in auditable sandboxes.

#7

REPLAY

Deterministic replay (physics-tight)

Same inputs + same environment → same hash

Replay manifest (WUManifestV1)
 • ociImageDigest (immutable)
 • inputRefId (CID + hash)
 • deterministicProfile (time, RNG, network)
 • canonicalizerId (normalize output)
 • expectedOutputHash

 Auditor submits ReplayReceiptV1 signed via EIP-712.

Hardening
 Sandbox untrusted code (microVM / gvisor).

 Require supply-chain attestations for images (SLSA / in-toto).

 Multiple auditors → hash quorum.

truth = [outputCanonicalHash, replay == outputCanonicalHash, id]

#8

EVALUATION

Evaluator modules

Deterministic grading at scale

Standard interface
 ITruthModuleV1.verify(wuid, evidence) → (truth, score, truthHash)

 EvaluatorReceiptV1 binds:
 • evaluatorCodeHash
 • submissionHash
 • score (fixed point)
 • evidenceCID (logs)

Common evaluators
 • unit tests / CI harness
 • schema + format checks
 • numerical tolerances
 • benchmark suites

 Aggregation: median(score) + threshold.

score_Bid1 = eval(score0, truth = (score_Bid1, 1))

#9

PROOF-FIRST

Proof-capable work templates

When proofs land, audits become a safety net

Template pins
 • programId / circuitHash
 • verifierGateway
 • public I/O schema hash
 • manifest schema hash
 • proofWindow + rebateBps

Public values
 Proof binds to:
 • manifestHash
 • outputCanonicalHash
 (+ optional score)

 No "proof of the wrong thing."

Economic upgrade
 Valid proof → audit premium rebate (or insurance)

 No proof → normal audit lottery sampling

 Gradient pushes system to proof-carrying labor.

Outcome: "validators vote" → "validators are continuously benchmarked by reality."

#10

MARKETS

Prover Market (sister pool to validators)

Truth throughput as a commodity utility

Stake-backed SLA
 Provers stake SAGIALPHA to offer capacity.

 Assignments lock stake until proof posted.

 Missed deadline → permissionless timeout → slashing + requeue.

Matching
 Reverse auction (MVP) → double auction (scale).

 Settlement is verifiable: on-chain checks proofs, not opinions.

 Delegated staking enables capital formation.

stake0Bps = n * [1/(unit, deadBps), time(Scale)]

#11

INCENTIVES

Validator staking tied to GSLI integrity

Reward honesty; slash divergence from truth

Audit-anchored Q(d)
 Let audit truth define a calibration signal:
 $Q(d) = \text{clamp}(\text{audit-truth agreement}, 0.8, 1.1)$
 This shifts integrity from social consensus → measured reality.

Reward split β
 $\beta_{\text{next}} = \beta_0 - Q(d)$
 • Validators earn more when quorum is diverse and accurate
 • Validators lose (slash) when they certify wrong outcomes.

 Peer pressure becomes mathematical.

$\beta_{\text{next}} = \beta_0 - Q(d)$

slash + divergence = stake + insurance

#12

Synthetic Labor Observatory (source)

Verbatim source slides (compressed).

VERIFICATION

Public verifiability surface

Every claim anchored to a URL + a hash

ENS subnames Role identities *.club / *.agent / *.node  app.ens.domains	AGIJobManager On-chain job primitive Contract source  etherscan.io	EIP-3668 CCIP-Read Offchain lookups  eips.ethereum.org
ENSIP-10 Wildcard resolution Namespace scaling  docs.ens.domains	Chainlink VRF Public randomness Audit lottery  docs.chainlink.com	deBridge SOL → ETH onboarding Cross-chain rails  docs.debridge.com

#13

OPERATIONS

Deployment (Green Path)

Deployable truth, reproducible by default

Day-one utility benchmark
 Run: make operator:green
 Pass criteria: CI banner shows  Day-One Utility and meets uplift guardrail.
 Archive outputs (JSON + HTML + PNG) as audit evidence.

One-click production
 Preflight: npm run deploy:checklist
 Deploy: npm run deploy:medick:auto ...
 Runtime: docker compose ... up -d
 Rollback: compose down → checkout prior tag → regenerate env.

CGreen + deploy@truth

#14

GOVERNANCE

Governance & safety posture

Antifragile operation with bounded blast radius

Owner controls Pause / resume Parameter limits Emergency timelocks Upgrade gates Early phases: reduce blast radius.	Transparency Open-source everything Reproducible builds Signed daily reports Immutable audit trails (CID + hash)	Autopoiesis Telemetry → incentives → adaptation Audit signals tune staking Proof-first lowers entropy System improves under stress.
---	---	--

#15

Build the Cathedral of Verifiable Work

Autonomy measured • Work proven • Value settled

Repos: AGIJobsv0 • AGI-Alpha-Agent-v0 • AGI-Alpha-Node-v0

#16

Global Synthetic Labor Index Prototype (source)

Verbatim source slides (compressed).



#0001

Executive Charter

Board / ministerial framing — identity → proof → settlement → governance

Identity

- Roles bound to keys (ENS / DID).
- Accountability by default.
- Scope rules prevent ambiguity.

Evidence

- Deterministic runs emit signed proof bundles.
- Replay beats rhetoric (audit as replay).
- Commit-reveal reduces bribery leverage.

Settlement

- Escrow releases only after validation.
- Receipts are settlement-grade artifacts.
- No payout without PoC.

Governance

- Policy gates, upgrades, emergency brakes.
- Operator override is final (bounded autonomy).
- MethodHash locks index rules per epoch.

#0001

Proof-of-Observability → Proof-of-Contribution

Observe → commit → attest → challenge → settle → index



Index-grade admissibility

- No PoC attestation → no index inclusion (and eventually: no payout).
- Validator signatures are economically liable (stake + slashing).
- Only store commitments on-chain; reveal plaintext only on dispute (privacy by default).
- Optimistic by default, adversarially correct via challenges.

Game theory (dominant strategy)

- Honest validators profit steadily.
- Dishonesty becomes negative-sum (slashing).
- Challengers are rewarded for accurate disputes.

#0002

α-Work Units (α-WU)

A canonical, auditable unit of verified synthetic labor

$$\alpha\text{-WU} = \text{VerifierGas}^{\theta} \times \text{Duration}^{(1-\theta)} \times \text{QualityScore}$$

$\theta \in [0,1]$ (start 0.5) • QualityScore $\in [0,1]$ (or 0.1/0.0000 normalized)

Metrology rules (v0)

- Count only PoC-attested settlements.
- Fails acceptance/SLO → 0 credit.
- Use robust stats (median, winsorize).
- Normalize on a rolling window (e.g., 13w).

Token coupling (thermostat)

Signals

- validated α-WU/epoch
- dispute rate
- SLO drift
- burn/emission ratios

Actuators

- fee splits
- burn fraction
- tier multipliers
- quorum + slashing

#0003

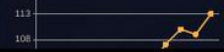
Global Synthetic Labor Index (GSLI) v0

Weekly macro observable: output × price × reliability × breadth

Constituents

- Q: verified output (Σ α-WU over PoC-attested settlements)
- P: clearing price (median AGIALPHA per α-WU)
- R: reliability (uptime + SLA − disputes)

Example (12-week sparkline)



#0003

Weekly Index Print (On-chain Anchor)

Public truth with private compute — Merkle-rooted datasets

Anchor fields (minimal)

```

weekID
datasetRoot // Merkle root of included jobs
m.ethodHash // frozen computation spec
indexValue
publishSetSig
challengeWindow
  
```

Audit posture

- Any auditor can recompute the print from datasetRoot.
- Outliers are winsorized; prices use medians.
- Weights locked for a quarter; changes bump methodHash.

Verifiable anchors (examples)

```

AG LHM anager:
ReB17806ba0685aaf908f7213588ec32fc1D5B4477
AG 3ALPHA del:
0x2e8f54c3ec41f5506c1082c081a099aa4e8e
M InterWauZ:
0x27d6e3868d852a2c26f4e820094903af8008
AG 3ALPHA v2:
0xAG1a3B3a130a9c28768EE8F97E21515A6846a19
Index inclusion rule: settled + PoC-attested + not
overturned
  
```

- deBridges onboarding: SOL, ETH (decimals + safe path)
- ENS roles: * stub.agileth / * agent.agileth / * node.agileth / * agileth
- Optional oracle mirror: Chainlink feed for GSLI prints

#0004

Global Synthetic Labor Index Prototype (source)

Verbatim source slides (compressed).

Synthetic Labor Yield Curve

Term structure of verified work (a-WW credits)

Instrument (v0): L2(T) labor zeros

Example curve (illustrative)

#0004

Labor-Backed Compute Notes

Finance expansion by selling long-dated verified output

Mechanism (simple path)



Risk controls

- Capacity-based issuance limit (quantile throughput + β).
- Performance bond + penalty buy-in on shortfall.
- Only external demand counts toward delivery (anti-wash).
- Tiered eligibility based on reliability history.

Macro flywheel

term structure + financing + more capacity + lower volatility + deeper curve + cheaper capital

- Proof-of-Contribution enforces delivery.
- Stashing + buy-in make default dominated.
- Index prints become the macro settlement reference.

#0005

Labor Repo Market

#0005

Standing Facility (Labor Open-Market Ops)

Corridor policy: cap stress, absorb excess liquidity

Two standing windows

- Ceiling: lend AGIALPHA vs SR/SRCP collateral at penalty rate.
- Floor: take AGIALPHA via reverse repo (escrow-collateralized) at deposit rate.
- Optional: bounded OMO buys/sells to shape the long end.

Policy corridor (illustrative)



Rule-bound policy (best practice)

- Set r^* from observables: labor inflation π_L , output growth g_Q , stress σ .
- Corridor widens automatically when σ rises (avoid facility dominance).
- Caps are hard limits; emergency actions only widen haircuts / pause new credit.
- Policy collateral: SR/SRCP (short-duration, overcollateralized).

#0006

Universal Inflation-Adjusted Credit Line

#0006

Invoices → Money-Market Instruments

Pool PoC-final receivables - mint SR/IR - finance SR in repo

Pooling constraints (money-market posture)

- Only PoC-final invoices (job settled; attestation survived challenge).
- WAM cap (e.g. 300d), WAL cap (e.g. 600d), concentration limits per borrower/cluster.
- Reserve account + junior retention (first loss).

Cashflow waterfall

- Repayments → fees + SR interest + SR principal + rebuild reserve + IR residual.
- Coverage tests (OC/IC/DQ) divert all cash to SR when stressed.
- IR retention aligns originator incentives (anti-adverse selection).

Tranches

- SR (Senior)
- IR (First-loss)
- SR aims to behave like short-duration "sash-like" paper.
- IR absorbs losses; originator retains minimum 5-15%.
- Wrap SR into SRCP_7D / SRCP_300 for clean repo collateral.

Repo financing

- Market makers finance SR/SRCP inventory in repo.
- Standing facility backstops SR/SRCP funding at the ceiling.
- Fixings (O/N, 7D, 30D) become the protocol short-rate complex.

#0007

Global Synthetic Labor Index Prototype (source)

Verbatim source slides (compressed).

SR/SRCP as Policy Collateral

A macro-stable balance sheet built from PoC-final economic events

Why SRCP works as "general collateral"

#0007

Publicly Verifiable Anchors

Contracts · ENS namespaces · onboarding links · standards

Ethereum contracts (Etherscan)

```
AG LickM anager
0x017886b5d066a5f908f7213588ec32fc1d5bA477

AG BULPHA treasury 6 decm a.k.
0x2e8f54c3ec41f5f06c1082c881a09ea4e0e

AG BULPHAequiM interWauk
0x27a6e8668d652A2c0Fad020D94903af98D8

AG BULPHA v2 0.8 decm a.k.
0xA61a383a138a9c20768EE8F97E21515A6846a19
```

Tip: always verify domain + contract + decimals before signing.

ENS roles + standards

```
Validators: * alpha.club.agi.eth | * club.agi.eth
Agents: * alpha.agent.agi.eth | * agent.agi.eth
Nodes: * alpha.node.agi.eth | * node.agi.eth
Sovereigns: * alpha.agi.eth | * agi.eth
```

Standards: EIP-712 (signature) · EIP-3668 (COP-Read)
ENSIPs: ENSIP-5 (text records) · ENSIP-10 (wildcards) · ENSIP-15 (normalize)

Onboarding: deBridge dePort (SQL) [ETH]
Oracle option: Chainlink mirror for index prints

#0008

Build the Cathedral of Verifiable Work

Autonomy measured · Work proven · Value settled

- 1) Prove work (PoC)
- 2) Print truth (GSLI)
- 3) Finance growth (curve → repo → corridor)

Repos: AGI.kbvs0 · AGI-Alpha-Agentv0 · AGI-Alpha-Nodev0

#0008

L0 Aegis / L1 Apotheosis Overview (source)

Verbatim source slides (compressed).

L0 Aegis and L1 Apotheosis Overview

Audit-complete programmable work · Adaptive governance · ENS-native identity

"As Columbus braved the seas, we now venture into the vast oceans of AGI... Dare to dream, dare to explore." — a-AGI Agent

AGI.Eth · AGI Jobs

Aegis (L0): conservation laws · Apotheosis (L1): adaptive control

#1

The Primitive: Programmable Work, Settled On-Chain

A minimal state machine that any observer can replay from logs.

Lifecycle (happy path)

1 Commit Job

Employer escrows value
+ job spec hash

2 Claim & Execute

Agent claims
+ produces artifacts

3 Validate

Validator committee
+ attests receipt

4 Settle

Contracts release / burn
+ finality & proofs

Design rule: if it cannot be replayed from events, it does not settle.

Aegis defines the ledger language. Apotheosis governs change without rewriting history.

#2

ENS Identity Lattice (Role-Native Authorization)

A low-entropy naming grammar that makes audits human-readable.

Namespaces (alpha and production)

Role	ENS Pattern	Meaning
Sovereigns	*.alpha.agi.eth *.agi.eth	Owner / guardians / businesses
Validators	*.alpha.club.agi.eth *.club.agi.eth	Attest receipts, earn fees, risk slashing
Agents	*.alpha.agent.agi.eth *.agent.agi.eth	Claim jobs, submit receipts, earn payouts
Nodes	*.alpha.node.agi.eth *.node.agi.eth	Provide compute, artifacts, observability

Registry-as-genome: stable semantics; resolvers for churn. Admission = ENS ownership + stake.

Names clarify actors unambiguously · No payout without validated proof · Official recognition is registry-driven

#3

L0 Aegis

The irreducible work ledger: commitments + receipts + settlement.

Autonomy measured · Work proven · Value settled

#4

L0 Aegis: Conservation Laws of Work

Small, stable contracts + golden events so anyone can replay truth.

The three sacred verbs

Commit (Work Intent)

Create ticketId + immutable params hash. Bind to employer identity and escrow.

Commit (Work Evidence)

Agent submits receiptHash / root referencing artifacts and proofs.

Settle (Conserved Value)

Escrow resolves exactly once: payout / burn / refund. Finality is evented.

Aegis invariants (must always hold)

- Conservation of value: escrowed = released + burned + refunded + locked.
- Single-finality: each ticketId settles exactly once (idempotent).
- Deterministic windows: disputes/claims are chain-time bounded.
- Receipt anti-replay: EIP-712 domain separation + nonce discipline.

If it is not reconstructible from events, it is not part of L0.

L0 stays frozen. Everything evolvable lives in L1 policy and tooling.

#5

Aegis Canon v1: Golden Events (Replay = Audit)

Design the logs so third parties can reconstruct state without trusting indexers.

Minimal event surface

JOB	RECEIPT	SETTLE
JobCommitted(ticketId, employer, paramsHash) - Creates canonical ticket - Binds to escrow - Binds to employer role identity	ReceiptCommitted / ReceiptAccepted - Agent evidence hash/root - Validator quorum attests - Windowed dispute semantics	FundsReleased / FundsBurned / Refunded - Conserved accounting - Idempotent finality - Always attributable to ticketId

Canon rule: every policy decision must be explainable as a function of (events, time, roles).

Aegis Canon v1 is the ABI/Log constitution integrations code against.

#6

L0 Aegis / L1 Apotheosis Overview (source)

Verbatim source slides (compressed).

SECTION

L1 Apotheosis

Evolvable policy, safety hooks, and operator control — without breaking L0.

Autonomy measured · Work proven · Value settled

#7

Apotheosis: Surgical Safety (Pause Lattice)

Contain incidents without freezing the entire economy.

Scopes

JOB	RECEIPTS	PAYOUTS	DISPUTES
Stop new commitments (contain contagion)	Stop new submissions (stop exploit propagation)	Stop fund egress (protect treasury)	Freeze adjudication (if validator set is compromised)

Control model

- Guardian: can pause scopes quickly; cannot drain funds or rewrite economics.
- Sovereign: multisig + timelock for policy upgrades and parameter changes.
- Every change emits diffs (ModuleUpdated / ParameterUpdated) for full history reconstruction.

Autopoeisis: the system reduces degrees of freedom under stress to preserve integrity.

#8

Policy Modules (Evolvable, Hot-Swappable)

Complexity lives in L1; L0 remains minimal and stable.

Module stack

RiskPolicy	eligibility, quorum, stake thresholds, routing
TaxPolicy	burn/withhold, fees, treasury flow
ReputationPolicy	scoring, decay, penalties, promotion alpha → open
DisputePolicy	validator set selection, evidence requirements

Constraint: policies may restrict behavior — but must never violate L0 invariants.

Governance is change with proofs: bounded, evented, and reversible where possible.

#9

One-Signature Onboarding (SOL → ETH)

A single user intent; relayed execution on Ethereum after bridge finality.

Flow (DLN hook → executor → mint v2)

User (Solana)	deBridge DLN	Onboarding Executor	AGIALPHA v2
Signs one order (bridge intent)	Fulfillment + EVM hook call	Receives transitory + calls vault	Minted to user (18 decimals)

Ethereum contracts (from one-pager v11)

Transitory \$AGIALPHA (6 decimals): 0x2e8fb543ec41f5f06c1f082c081a609eaa4e4be
 AGIALPHAEqualMinterVault: 0x27d0f8668c6f652a26f4e020d94903af80d8
 Final \$AGIALPHA v2 (18 decimals): 0xA61a383a130a9c20768EBF97E21515A6046a11A

Best practice: hook execution is optional success with fallback delivery if the executor is paused.

#10

One-Signature: Capital + Identity + Capability

Bridge → Mint v2 → Mint ENS role surname → Stake/Register — in one intent.

Atomic onboarding payload

Hook target executes:

- Mint AGIALPHA v2 to executor (so it can pay fees/stake without extra approvals).
- Create ENS surname under role parent node (e.g., alpha.agent.agi.eth).
- Set resolver addr record → user address; transfer wrapped name to user.
- Stake required amount; register role in CapabilityRegistry.
- Sweep remaining v2 to user.

Outcome: user arrives named, categorized, and able to act.

Role parents

Validators: *alpha.club.agi.eth | *club.agi.eth
 Agents: *alpha.agent.agi.eth | *agent.agi.eth
 Nodes: *alpha.node.agi.eth | *node.agi.eth
 Sovereigns: *alpha.agi.eth
(Note: using alpha, keeps l0 rules minimal and avoids irreversible locks unless required.)

This collapses activation energy: one signature → ready-to-work identity with bonded credibility.

#11

Replace Allowlists with Physics

Eligibility becomes an on-chain function of ENS ownership + bonded stake.

Admission control Hamiltonian

$$H(a) = \lambda_1 1[-nameOK(a)] + \lambda_2 1[stake(a) < S_{req}] + \lambda_3 1[expired(a)]$$

Admit iff $H(a) = 0$

nameOK(a)

- a controls a role surname under the correct parent (agent/club/node).
- If wrapped: ownerOf(node) == a and expiry > now.
- If unwrapped legacy: ENS.owner(node) == a (during migration).

stake(a) ≥ Sreq

- Stake held in CapabilityRegistry with withdrawal delay.
- Sreq scales with job value & system temperature.
- Slashing applies only for provable faults.

Dominant strategy: the honest path is cheapest; Sybil scale requires proportional locked capital.

#12

L0 Aegis / L1 Apotheosis Overview (source)

Verbatim source slides (compressed).

Validation as a Coordination Game

Stake-weighted committees + accept-duty bonds + provable slashing.

Committee assignment (per job)

Algorithm (scalable)

- Eligible set = owns validator name + stake $\geq$ threshold.
- Build weight index (Fenwick/segment tree) for $O(\log N)$ sampling.
- Sample k validators with probability $\propto$ stake (cap optional).
- Store committee on-chain; vote window begins.

Duty acceptance (anti-grief): validators post a bond only when they acceptDuty(jobId).

Randomness tiers

Tier 0 (routine jobs)

Delayed blockhash / PREVRANDAO mix — acceptable when value is low.

Tier 1 (high value)

Chainlink VRF v2.5 — stronger unpredictability; pay in LINK or native.

Tip: PREVRANDAO is biasable/predictable; reserve it for low-stakes tiers.

Committee validation makes honesty cheap and collusion expensive at scale.

#13

Slashing Hooks: Only What You Can Prove

Avoid governance wars: slash for cryptographic faults and accepted-duty liveness only.

Fault classes

SAFETY

Provable by signatures

- Double-sign: same validator signs conflicting votes for one job.
- Conflicting approvals: approves two receipt hashes for the same job.
- Replay protection: EIP-712 domain separator + per-validator nonce.
- Reporter bounty from slash makes monitoring incentive-compatible.

LIVENESS

Provable by on-chain state

- No acceptDuty → no liveness slashing (prevents grieving).
- acceptDuty locks bond; voting before deadline unlocks.
- Accepted but missed deadline → bond slashed automatically.
- Bonds and deadlines scale with job value and temperature.

Slash rarely, prove always: safety and liveness are separate fault domains.

#14

Adaptive “Temperature” Controller

Automatically raise barriers under attack; relax when stable — bounded and auditable.

Epoch metrics → parameter updates

$p = \text{disputeRate}$, $p90 = \text{payout bucket percentile estimate (histogram)}$

What increases with T

- Required stake multiplier
- Committee size k
- Duty bond rate
- VRF-tier threshold (more jobs use VRF)

Safety rails (must hold)

- Hard bounds for each parameter
- Per-epoch change caps (anti-oscillation)
- Dispute bonds to reduce gaming
- Guardian can pause controller

Control theory: tune the system like a thermostat, not a brittle rulebook.

#15

Public Ledger Explorer (Event-Sourced Truth)

A third party can replay jobs, receipts, and payouts directly from logs.

Replay pipeline



Why it matters

- Independent verifiers can reproduce accounting and settlement from events alone.
- ENS role names make disputes and reputation human-legible.
- Evidence archives link to receipts, validator attestations, and policy versions.

Audit becomes a fusion: replay, compare, export.

#16

Institutional Ops: Stay Green

Reproducible demos, safety checklists, and audit artifacts — from day one.

Runbook (from AGI Jobs one-pager v11)

Pilot / Green Path

`make operatorgreen`

- Pass criteria: banner starts with Day-One Utility; meets uplift guardrail.
- Review artifacts: report JSON, HTML dashboard, PNG snapshot, owner controls snapshot.
- Pause/resume: make owner-toggle - Reset defaults: make owner-reset.

Production deploy (one-click)

- `rpm run deploy:checklist`
- `rpm run deploy:onedclick:auto -- --config ... --network ... --compose`
- `docker compose ... up --build -d`
- Rollback: compose down → checkout tag → regenerate env from archived manifest

Rule: treat CI-green as deployable truth; archive evidence with release tags.

Institutional reliability is a habit: reproducibility, least privilege, and provable change history.

#17

Shipset (next commits)

- Aegis Canon v1: frozen ABIs + golden events + invariant test suite
- One-Signature Onboarding: DLN hook → mint v2 → ENS role subname → stake/register
- Public Ledger Explorer: event-sourced replay + ENS identity resolution + proof exports
- Validator Game: stake-weighted committees + EIP-712 votes + provable slashing + thermostat

Key on-chain anchors

AGIJobManager: 0x0178B6baD06aaF908F7213588eC32F105bA477
 Transitory AGIALPHA: 0x2e8fb54c3ec41f53706c1f082c081a609eaa4e4be
 MatterVault: 0x216EE86684652A260F4E020949803aF80D9
 Final AGIALPHA v2: 0xA61a3B3a130a9c2076EEBF97E21515A6046a1fA

Autonomy measured · Work proven · Value settled

#18

Choosing the First Lineage Asset (source)

Verbatim source slides (compressed).

Choosing the First Lineage Asset

Institutional decision brief

Genesis Charter → ENS Grammar → Shadow-Fork Policy Loop

v1 - owner-gated autonomy

AGLEth / AG1 Jobs v0

#1

Executive Charter

Identity

- Roles bound to keys (ENS/DID)
- Accountability by default

Evidence

- Deterministic runs emit signed proofs
- Replay beats rhetoric (audit as replay)

Settlement

- Escrow releases only after validation
- Receipts are settlement-grade artifacts

Governance

- Policy gates, upgrades, emergency brakes
- Operator override is final (bounded autonomy)

No value without evidence · No autonomy without authority · No settlement without validation

#2

What Is a Lineage Asset?

Definition

- Names the system (canonical Schelling point)
- Binds authority (who can do what, under which brakes)
- Spawns descendants (validators / agents / nodes / sovereigns)
- Persists across upgrades (audit continuity)

Selection criteria (best practice)

- Unambiguous identity
- Transfer is a ceremony (not accidental)
- Policy-bounded execution
- Deterministic audit trail
- Clear alpha → main promotion path

#3

Decision: First Lineage Asset

First lineage asset for alpha → main

Sovereign Genesis Charter

ENS-anchored · multisig-controlled · manifest-pinned · transfer-constrained

Canonical names

- Alpha epoch
- genesis.alpha.agi.eth (root)
- alpha.agi.eth (membrane)
- Promotion to main
- genesis.agi.eth (mirrors manifest lineage)
- mainline policy registry · allowlists

Why this wins (game theory)

- Collapses ambiguity (Schelling point)
- Upgrades remain legible (continuity)
- Role issuance is deterministic
- Recovery is predefined
- Forks retain provenance

Measured control objective

$$F = -E[R] + \lambda[w - w_{prior}]^2 + \text{penalty}(\text{guardrail violations})$$

Reward R aggregates: risk1 · liveness1 · reversibility1 · cost1

Execution remains owner-signed.

#4

Genesis Charter: What the Root Must Publish

Control plane

- Controller = multisig Safe (threshold > 1)
- Timelock for irreversible ops (upgrades, treasury)
- Break-glass pause allowed fast; postmortem required
- Key custody + rotation ceremony

Manifest (hash-pinned)

```
{
  "lineage_id": "...",
  "epoch": "alpha",
  "root_ens": "genesis.alpha.agi.eth",
  "owner_multisig": "0x...",
  "role_namespaces": {
    "validators": "...alpha.club.agi.eth",
    "agents": "...alpha.agent.agi.eth",
    "nodes": "...alpha.node.agi.eth",
    "sovereigns": "...alpha.agi.eth"
  },
  "policy_hash": "keccak256(...)"
}
```

Transfer constraints

- Transfer is a ceremony, not an accident
- Registrar/controller held by multisig
- Controller changes are timelocked
- Public announcement + signed audit bundle
- Recovery playbook tested in shadow mode

Outcome: continuity across upgrades and custody events.

#5

ENS Grammar: One Rule + Scoped Meaning

Canonical form

<entity>.{<env>}<role>.agi.eth

role ∈ {club, agent, node} · env ∈ ENV_SET (optional)

Examples by role

Validators (club)

alice.club.agi.eth | alice.alpha.club.agi.eth

stake-bonded verification · slashable accountability

Agents

helper.agent.agi.eth | helper.alpha.agent.agi.eth

work execution · proofs · job claims

Nodes

gpu01.node.agi.eth | gpu01.alpha.node.agi.eth

deterministic runtime · metering · observability

Guardrails: env-scoped names are developed only within env.agi.eth · Reserve (agent,node.club) under every env.agi.eth

#6

Choosing the First Lineage Asset (source)

Verbatim source slides (compressed).

Registry-as-Genome

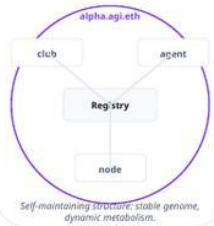
Autopoietic control loop

- Membrane: env.agi.eth defines the boundary
- Organs: role roots (agent / node / club) separate duties
- Genome: registry publishes ENV_SET + canonical packages
- Metabolism: resolvers turn names into live endpoints
- Immune system: status + proofs + slashing correct drift

Invariants

- Names classify actors unambiguously (role suffixes stay meaningful)
- No payout without validated proof; no settlement without validation
- Official recognition is registry-driven (not self-asserted)
- Push churn into resolvers (wildcards / CCIP-Read), not naming

Cell model (env as membrane)



agi.eth

7/15

#7

Shadow-Fork Digital Twin

Mirror production

- Prometheus time-series snapshots + SLOs
- Alert history (open/close)
- On-chain events (pause/unpause, setValidator, ...)

Score outcomes

- Reduce risk
- Preserve liveness
- Remain reversible
- Minimize cost

Policy tuning: minimize measured free energy over historical incidents
Deploy in shadow first · Execute only with owner signature

Shadow-mode loop (no execution)



agi.eth

8/15

#8

Incident-Jobs Loop



Agents (propose)

- Read incident bundle (alerts + metrics + chain events)
- Generate candidates (allowed action catalog)
- Simulate on fork; compute outcome deltas
- Emit proposal artifact + proof bundle (hash-addressed)

Validators (verify)

- Replay simulation at incident fork block
- Verify state diffs + scoring
- Attest / dispute (commit-reveal); slashing enforces honesty
- Output: settlement-grade validation receipt

Simple path = dominant strategy: replayable evidence beats narrative.

agi.eth

9/15

#9

Minimal Wiring Plan (Plug In Tomorrow)

1) Allowed action catalog

- Addresses + ABIs + selectors
- Role-gated (owner/admin/timelock)
- Prechecks + caps
- No arbitrary external calls

2) AnvilOutcomeExtractor

- Read S0 → apply tx → read S1
- Compute deltas (risk/liveness/reversible/cost)
- Feed scoring + learning

3) Incident store + audit

- SQLite for queryability
- Append-only JSONL for signing
- Artifact hashes/CIDs for replay packs
- Nightly: policy diff + shadow vs owner

4) Shadow-run daemon

- Alert opens → incident @ fork block
- Run policy in shadow (no signing)
- Propose tx bundle + evidence
- Alert closes → finalize + label

agi.eth

10/15

#10

One-Signature Onboarding

UX primitive: an Onboarding Intent (one signature) → replay-protected, executed by keepers / paymasters

Flow A — Start on Solana

- User signs one Solana bridge tx
- Keeper executes destination claim
- CallProxy triggers OnboarderRouter.onboard(...)
- Router executes: mint/claim → ENS subname → role registry → Onboarded(intentHash)

Flow B — Start on EVM (AA)

- User signs one ERC-4337 UserOperation
- Paymaster sponsors gas (strict allowlist)
- Smart account executes onboarding multicall

Best-practice guardrails

- Exact calldata allowlist
 - Per-user caps
 - Mandatory simulation
 - Replay protection (intentHash/nonces)
- Destination outcome: user owns a role identity [e.g., alice.alpha.agent.agi.eth · guu01.alpha.node.agi.eth]

One human intent. One signature. Many deterministic steps. Full audit trail.

agi.eth

11/15

#11

Session Keys for Agents

Capability delegation (bounded)

- Grant time-boxed, scope-boxed session key to agent runtime
- TTL: 24h (hot) / 7d (warm)
- Scopes: post jobs · claim jobs · submit completion
- Hard-blocked: withdraw treasury · upgrade · arbitrary calls
- Revocation: single on-chain revoke (+ global kill-switch)

Architecture

- Agent identity is a smart account with SessionKeyValidator
- Policy/Registry defines allowed targets + selectors
- Validator enforces scope + quotas + TTL
- Telemetry emits SessionEnabled/Used/Revoked events

Default capability template (v1)

Scope	Allowed
POST_JOB	AGIJobManager.postJob(...) blocks token drain / arbitrary approve
CLAIM_JOB	AGIJobManager.claimJob(jobId) quota prevents flooding
SUBMIT	AGIJobManager.submitCompletion(...) blocks overwrite of others
ADMIN	None withdraw/upgrades/setOwner are owner only

Telemetry: SessionEnabled · SessionUsed · SessionRevoked → feeds the incident loop

agi.eth

12/15

#12

Choosing the First Lineage Asset (source)

Verbatim source slides (compressed).

The Lineage Ceremony

Seven steps (alpha)

1. Create sovereign multisig (hardware-backed keys)
2. Assign genesis.alpha.agi.eth controller to multisig
3. Publish Genesis Manifest hash (ENS + immutable storage)
4. Bind env registry + role registrars (club/agent/node)
5. Issue first credentials (1-3 validators, agents, nodes)
6. Run Green Path; archive audit artifacts
7. Freeze epoch: tag manifest + sign release bundle

Avoid

- Token first, constitution later
- Unbounded admin keys
- Role minting before root-of-trust
- Irreversible locks before migration plan

Promotion to main (exit criteria)

- Proofs reproducible
- Brakes tested
- Incident loop stable
- Upgrade paths audited
- Custody rotations rehearsed

agi.eth

13/15

#13

Metrics, Guardrails, and Proof

Outcome metrics (per incident)

- risk_delta (↓ is good)
- liveness_delta (↑ is good)
- reversibility (binary → time to rollback)
- cost (gas + externalities)
- operator_time_saved

Goal: throughput with bounded risk

Scoring (example)

$$R = w_r \text{risk} + w_l \text{liveness} + w_{rev} \text{reversibility} - w_c \text{cost}$$

Artifacts required: (S0, tx, S1, deltas, calldata, gas, signatures, replay scripts) : If it cannot be replayed, it does not govern.

Guardrails (fail-closed)

- Never reduce liveness below floor without explicit risk justification
- Disallow non-reversible actions except via timelock
- Require simulation receipts for any proposed tx
- Owner signature remains the execution bottleneck

agi.eth

14/15

#14

Build the Cathedral of Lineage and Verifiable Work

Autonomy measured · Work proven · Authority preserved

Artifacts: [genesis](#) [manifest](#) [proofs](#) [value](#) [scripts](#)

Next: incident-jobs → session keys → one-click sovereign ops

Repos: [AGIJobsV0](#) · [AGI-Alpha-Agent-v0](#) · [AGI-Alpha-Node-v0](#)

agi.eth

15/15

#15

AGI Jobs — Epoch Snapshot Market v1 (source)

Verbatim source slides (compressed).

AGI Jobs Epoch Snapshot Market v1

From procurement auctions → continuous clearing → ERC-1155 coverage futures

INSTITUTIONAL v1 LOW-ENTROPY LIVENESS GASLESS UX

#1

Problem

Per-epoch procurement auctions do not scale with participation.

Why discrete procurement breaks at scale

Failure modes

- Volatility: thin bids → spiky prices, unpredictable cost curves.
- Liveness cliffs: no winner / winner fails → epoch stalls.
- Wasted compute: many bidders prove; one gets paid.
- Paymaster stress: emergency retries amplify sponsored gas burn.

Design target

- Constant-cost verification onchain; heavy work offchain.
- Predictable coverage inventory over a rolling horizon.
- Backups + rescue paths make liveness a guarantee, not hope.
- Validators: one sponsored action → attested event + receipt.

AGI jobs - Epoch Snapshot Market v1

#2

Principle

Reduce uncertainty and cost via deterministic, replayable settlement.

Build a low-entropy coordination loop

Gibbs free energy framing (operational)

$$F = U - TS$$

U: economic utility captured • S: ambiguity / dispute surface • T: operational friction

Goal: drive T·S → 0 via determinism + reproducible receipts + market incentives.

Determinism

- One canonical JobSpec
- Digest-bound VSA
- Replayable verification

Incentives

- Stake + audits
- Backups paid to rescue
- Slashing dominates cheating

Inventory

- Rolling coverage buffer
- Standing offers
- No panic buying

AGI jobs - Epoch Snapshot Market v1

#3

Architecture

A deterministic arc: discrete → continuous → tokenized coverage.

The evolution arc

Procurement (epoch auction)

- Commit/reveal bids
- Single winner
- High volatility

Continuous clearing

- Standing offers
- Best-K stored
- Rolling buffer

Coverage futures (ERC-1155)

- Primary + backups
- Tradable obligations
- Margin & slashing

Capacity vaults (ERC-4626)

- Pool risk
- Smooth yield
- Guarantee inventory

Key move: treat liveness as inventory, not bidding luck.

AGI jobs - Epoch Snapshot Market v1

#4

Pipeline

Join intents → batch proof → epoch root → cheap membership checks.

Epoch snapshots: the admission macrostate

Join intents

validators → JoinBook

Batch proof

zkVM checks ENS ownership

Publish epoch

store root + metaDigest

Attest

Merkle gate + VSA

Onchain surface (minimal)

- EpochAdmission: (epochId → IHHash, root, metaDigest, leafCount).
- TournamentHub: attest only if membership proof passes + validator signature valid.
- Paymaster policy: sponsor attest path; reject anything that can revert.

Why it scales

- One proof per epoch (amortized), not per validator.
- Membership checks are O(log N) Merkle proofs.
- Epoch bundles are content-addressed; availability enforced by challenges.

AGI jobs - Epoch Snapshot Market v1

#5

Market

Standing offers + rolling buffer + stable pricing + high liveness.

Continuous clearing (no more per-epoch panic)

State the protocol maintains

Coverage buffer over the next H epochs: primary[E] + backups[E][..]

Offers are continuous; assignments lock at E - LOCK_AHEAD (damping).

Order book (bounded)

- Per epoch store best-K offers only
- Tie-break by SLA + bond quality
- Deterministic selection

Clearing

- Primary = best offer
- Backup = next offer(s)
- Dutch rescue if coverage missing
- Second-price payout (stability)

AGI jobs - Epoch Snapshot Market v1

#6

AGI Jobs — Epoch Snapshot Market v1 (source)

Verbatim source slides (compressed).

Coverage Futures

Tokenize delivery obligations; let capacity move without panic.

ERC-1155 coverage slots

Slot = tradable obligation to deliver epoch snapshot (root + proof + metaDigest)
 tokenId = keccak256(chainId, epochId, lane)

Lanes

- PRIMARY: must deliver by deadline; earns capacity premium.
- BACKUP: may deliver on primary miss; paid from slashed margin.
- Optional BACKUP2 for higher safety at scale.
- Anti-theft: proof binds to slotOwner (public input).

Transfer & delegation

- Two-step transfer: propose → accept + rebond.
- Lockout window near delivery (reduces last-minute chaos).
- Liquidation: if bond insufficient, auction the slot.

AGI jobs - Epoch Snapshot Market v1

#7

Economics

Two-part tariff + second-price payout + deterministic SLAs.

Pricing, SLA, and settlement

$$\text{Payout}(E) = \text{fixedFee}(E) + p_{\text{leaf}} \cdot \text{leafCount}(E) + \text{SLA_bonus}(E)$$

fixedFee cleared by market; p_leaf updated slowly (thermostat) to keep delivery success high.

SLA-A

Fast deliver

Multiplier: 1.15x

- Earlier deadline
- Higher bond
- Higher reward

SLA-B

Standard

Multiplier: 1.00x

- Earlier deadline
- Higher bond
- Higher reward

SLA-C

Best-effort

Multiplier: 0.85x

- Earlier deadline
- Higher bond
- Higher reward

Second-price payout: winner paid at next-best fixedFee (reduces bid shading).

AGI jobs - Epoch Snapshot Market v1

#8

Safety & Liveness

Backups + slashing + rescue paths turn liveness into a guarantee.

Liveness guarantees

Deterministic safety rails

- Bonded margin on each slot (fixed + optional per-leaf).
- Miss deadline → slash; backup paid to deliver.
- Invalid proof → full slash (objective).
- Data availability enforced via challenge-response.

Rescue mode (never stall)

- If coverage missing at lock: Dutch rescue reward ramps until claimed.
- Claim requires high bond; failure is expensive.
- Optional shadow publisher for high-value epochs.

AGI jobs - Epoch Snapshot Market v1

#9

Validator UX

One sponsored action → onchain event + auditable receipt bundle.

Validator path: gasless and deterministic

Single action flow

Input

- Validator uses ENS-shaped identity (namespace-gated).
- Gets membership proof for current epoch (Merkle).
- Signs VSA for a job submission.
- Sends ERC-4337 UserOp (paymaster sponsors).

Output

- Onchain Attested(...) event emitted.
- Artifacts written: jobspec.json, run-report.json, vsa.json.
- Indexers update leaderboard + reliability scores.

Admission ticket: *alpha.club.agi.eth | *club.agi.eth

AGI jobs - Epoch Snapshot Market v1

#10

Paymaster

Sponsor only what is safe, bounded, and non-reverting.

Paymaster rules (budget scales)

Allowlist & checks

- Allowlist contract + selector: TournamentHub.attest(...) only.
- Staticcall verifyLeaf(epochId, leaf, proof) before sponsoring.
- Bound calldata size and refuse unknown targets.
- Per-user + global budgets; epoch-level rate limits.

Why it works

- Membership checks are cheap and deterministic.
- Attack surface collapses: no external calls needed to sponsor.
- Amortization: epoch snapshot proof is paid once, reused by many.

AGI jobs - Epoch Snapshot Market v1

#11

Proof Compression

Constant-size SNARKs keep onchain verification predictable.

Succinct proofs for massive throughput

Key move

Replace large L1 state proofs with a constant-size SNARK that binds to the L1 anchor.

Onchain verifier

- Verify constant-size proof
- Store (epochId → root)
- Emit EpochPublished(...)

OP Stack anchor: L1Block predeploy at 0x4208...0019

AGI jobs - Epoch Snapshot Market v1

#12

Throughput result

- One proof per epoch, verified once.
- Validators use cheap Merkle proofs.
- Paymaster spend scales with real usage.

Example: Geth16 proofs ~360 bytes; ~270k gas verify (doc values).

AGI Jobs — Epoch Snapshot Market v1 (source)

Verbatim source slides (compressed).

Risk Pooling

Vaults turn spiky obligations into smooth capacity yield curves.

Proof capacity vaults (optional vNext)

Vault idea

Pool collateral + capacity across many publishers; sell slots forward; socialize variance.

Mechanics

- Deposit collateral → shares
- Vault posts offers & mints slots
- Payouts accrue; losses pooled
- Inventory target auto-maintained

Benefits

- Smooth yield for capacity providers
- Lower per-slot overcollateralization
- Higher liveness for same spend

AGI jobs - Epoch Snapshot Market v1

#13

Next

Ship the conformance loop, then scale liveness with slot futures.

Immediate ship list

- Deploy EpochAdmission + TournamentHub + paymaster allowlist on a cheap OP-Stack L2.
- Run epoch snapshot publisher with backup coverage; publish metaDigest bundles.
- Introduce continuous clearing with best-K offers + rolling horizon buffer.
- Mint ERC-1155 slot futures for primary/backup lanes; enforce margin + slashing.
- Add Dutch rescue + challenge-response data availability for "never stall".

ENS namespaces: validators *.alpha.club.agi.eth | *.club.agi.eth • agents *.alpha.agent.agi.eth | *.agent.agi.eth • nodes *.alpha.node.agi.eth | *.node.agi.eth

AGI jobs - Epoch Snapshot Market v1 - 2025

#14

Unified Macro Stack (24-slide source)

Verbatim source slides (compressed).

Global Synthetic Labor

Unified Macro Stack — index-grade evidence → identity safety → markets → policy

L0 Aegis + L1 Apotheosis → AGI Passport (EIP-7702) → Synthetic Labor Observatory → GSLI → Macro Facilities

Autonomy measured • Work proven • Value settled

#1

Executive Charter

Identity

- Roles bound to keys (ENS/DID).
- Accountability by default.

Evidence

- Deterministic runs emit signed proof bundles.
- Replay beats rhetoric (audit as replay).

Settlement

- Escrow releases only after validation.
- Receipts are settlement-grade artifacts.

Governance

- Policy gates, upgrades, emergency brakes.
- Operator override is final (bounded autonomy).

No value without evidence • No autonomy without authority • No settlement without validation

#2

SECTION 1

Ledger & Governance

Freeze truth at L0. Evolve policy at L1 — without rewriting history.

#3

L0 Aegis + L1 Apotheosis

Primitive: programmable work settled on-chain

Design rule

- Replayable state machine: any observer reconstructs truth from events.
- If it cannot be replayed from events, it does not settle.
- L0 defines conservation laws; L1 adapts parameters under explicit brakes.

Lifecycle (happy path)

- | Step | Action | Output |
|------|-----------------|-------------------------|
| 1 | Commit Job | escrow + paramsHash |
| 2 | Claim & Execute | artifacts + logs |
| 3 | Validate | committee attests |
| 4 | Settle | release / burn / refund |

#4

Aegis Canon v1 — Golden Events

Conservation laws (must always hold)

- Conservation of value: escrowed + released + burned + refunded + locked.
- Single finality: each ticketID settles exactly once (idempotent).
- Deterministic windows: disputes/claims are chain-time bounded.
- Receipt anti-replay: EIP-712 domain separation + nonce discipline.
- Role legitimacy: actor ENS identity is resolvable and evented.

Minimal event surface

JOB

JobCommitted(ticketId, employee, paramsHash)

RECEIPT

ReceiptCommitted / ReceiptAccepted

SETTLE

FundsReleased / FundsBurned / Refunded

Canon rule: Every policy decision must be explainable as a function of (events, time, roles).

#5

First Lineage Asset — Genesis Charter

Schelling point for truth + authority

- Names the system (canonical root) and persists across upgrades.
- Binds authority: who can do what, under which brakes.
- Spawns descendants (validators / agents / nodes / sovereigns).
- Transfer is a ceremony (timelocked + audited), not an accident.
- Shadow mode learns; owner signature executes.

Measured control objective (free-energy proxy)

$$F = -E/R + \lambda [w - w_{\text{prior}}]^2 + \text{penalty}(\text{guardrail violations})$$

R aggregates: risk, liveness, T, reversibility, T, cost, l

Genesis Manifest (hash-pinned) — minimum fields

```
{
  "epoch": "alpha",
  "root": "genesis.alpha.agieth",
  "timestamp": "2024-01-01T00:00:00Z",
  "to_krum_spaces": {
    "validators": "alpha.club.agieth",
    "agents": "alpha.agents.agieth",
    "nodes": "alpha.nodes.agieth"
  },
  "policy_hash": "keccak256(L, F)"
}
```

#6

Unified Macro Stack (24-slide source)

Verbatim source slides (compressed).

SECTION 2

Identity & Safety

One signature, bounded blast radius: fuse membranes + witnessed delegation + recovery.

#7

Role Lattice (ENS Namespaces)

Namespace grammar

- `<entity>(<env>)<role>agi.eth`
- role ∈ {club, agent, node}, env optional (e.g., alpha).
- Guardrail: reserve (agent,node,club) under every env.agi.eth.

Roles

Validators

*alpha.club.agi.eth | *club.agi.eth
Witness STHs, attest receipts, slashable.

Agents

*alpha.agent.agi.eth | *agent.agi.eth
Execute work, sign outgoing txns, principles.

Nodes

*alpha.node.agi.eth | *node.agi.eth
Run jobs, emit telemetry, bonded.

Sovereigns

*alpha.agi.eth | *agi.eth
Root-of-trust keys, policy roots, DTLL.

Registry-as-Genome (autopoiesis)

- Membrane: env.agi.eth defines the boundary.
- Genome: registry publishes ENV, SET + canonical packages + recognizedAliases.
- Metabolism: resolvers/gateways handle churn, anchors stay stable.
- Immune system: status + proofs + slashing detect and correct drift.

#8

AGI Passport — Civilization-Grade Identity

Design goal: one approval, bounded blast radius

Identity Membrane

- ENS names + fuse presets create an unforgeable role identity.
- What can change is explicit; what cannot is fuse-locked.

Delegation Transparency

- Every EIP-7702 delegation becomes an auditable event.
- Merkle logs + witnessed STHs make state shared reality.

Auto-Healing Recovery

- Pre-signed kill-switch + quarantine delegation.
- Recovery works even if the user is offline.

Passport kernel

- ENS Name + Authority (EIP-7702/AA) + Policy + Attestations + Commitments (stake/bonds).
- Routing invariant: No Verified Delegation → No Sponsored Gas → No Job Routing → No L2 Activation.

#9

One-Signature Onboarding + ENS Fuse Membranes

One intent. Many deterministic steps. Zero ambiguity.

- Batch: bridge onboarding (where possible) + mint/claim + ENS subname + role registry.
- Outcome: constrained action space + auditable delegation record + recovery path.

Representative ENS fuses (role preset)

- CANNOT_UNWRAP — prevents bypass of wrapper restrictions.
- CANNOT_TRANSFER — identity cannot be sold or stolen by transfer.
- CANNOT_SET_RESOLVER — prevents stealth record redirection.
- PARENT_CANNOT_CONTROL — stronger child sovereignty.

Onboarding sequence (illustrative)

User approves
EIP-7702 auth
↓
Verified
Delegator
↓
Batch ENS
+ role + fuses
↓
Commit stake
+ keys
↓
Mirror L2
activation

#10

EIP-7702 Safety Layer — DTL + DTLL + Recovery

Threat model (if ungoverned)

- Phished authorization installs malicious execution.
- Upgradable delegates can time-bomb an identity.
- Replay surface expands if authorizations are chain-agnostic.

Wallet enforcement rule

- No witnessed transparency → no delegation UX.
- Require inclusion proof to latest witnessed STH.
- Quarantine on anomalies; recover via kill-switch.

Controls (dominant strategy)

- Allowlisted delegates (codehash pinned).
- DTL transparency logs + DTLL signed log lists (x2 independent logs).
- Rescue: Seed Vault kill-switch + sterile quarantine delegation.

Operational loop

Prevent
Fuses +
allowlist
+
Least
privilege

Detect
DTL/
DTLL
Witness
ed STHs

Recover
Seed
Vault
SD
quaranti
ne

#11

SECTION 3

Evidence & Observatory

Proof-of-Contribution turns telemetry into economic evidence. The index becomes auditable truth.

#12

Unified Macro Stack (24-slide source)

Verbatim source slides (compressed).

Proof-of-Observability → Proof-of-Contribution

Lifecycle

- Observe → Commit → Attest → Challenge → Settle → Index
- Index-grade admissibility: no PoC attestation → no index inclusion.
- Privacy by default: commitments on-chain; reveal plaintext on dispute.

Game theory (dominant strategy)

- Honest validators profit steadily; dishonesty becomes negative-sum (slashing).
- Challengers are rewarded for correct disputes.
- If it cannot be replayed, it does not settle.

Verification Summary Attestation (EEP-712)

```

struct VSA {
  address jobManager;
  uint256 jobId;
  address worker;
  address validator;
  bytes32 jobSpecHash;
  bytes32 outputHash;
  bytes32 outputMerkleRoot;
  uint256 alphaU;
  uint256 reliabilityScore;
  uint64 deadline;
  uint256 nonce;
}

```

agp.eth 18/24

#13

α-WU Metrology (Verified Work Units)

Canonical, auditable unit of verified synthetic labor

- Count only PoC-attested settlements.
- Fails acceptance/SLO → 0 credit.
- Robust stats: median, winsorize extremes, rolling normalization.

Definition (v0)

$$\alpha\text{-WU} = \text{VerifierGas}^{\delta} \times \text{Duration}^{(1-\delta)} \times \text{QualityScore}$$

$\delta \in [0, 1]$ (start 0.5) - QualityScore normalized

Four reproducible KPIs (from public event logs)

- Acceptance (pass/fail, auditable).
- Quality (stake-weighted score).
- On-time (SLO adherence).
- Yield (slashing-adjusted).

agp.eth 14/24

#14

GSLI — Global Synthetic Labor Index (v0)

Weekly macro observable: output × price × reliability × breadth

- Q: L α-WU (PoC-attested)
- P: median AGALPHA / α-WU
- R: uptime/SLO — disputes
- B: entropy/diversity

Index level (weekly)

$$\text{GSLI}_t = 100 \cdot \exp \left(0.55 \Delta \ln Q + 0.25 \Delta \ln P + 0.15 \Delta \ln R + 0.05 \Delta \ln B \right)$$

Weights locked per quarter; changes bump method hash.

Weekly Index Print (on-chain anchor)

- weekId, datasetRoot (Merkle of included jobs)
- methodHash (frozen computation spec)
- indexValue, publisherSig, challengeWindow
- attestation coverage + disputes published each week
- Anyone can recompute from datasetRoot.

agp.eth 15/24

#15

Integrity Reserve + Truth Modules

Integrity Reserve

- Every work unit pays a tiny audit premium.
- Public RNG selects WUs for deep audit (audit lottery).
- Auditors paid via proper scoring rules against revealed truth.
- Benchmark validators against reality — continuously.

Truth Modules (physics-tight)

- Deterministic replay: same inputs + same environment → same hash.
- Evaluator harness: deterministic score for pass/fail + continuous score.
- Proof-carrying work: zkVM/SNARK binds manifestHash + outputCanonHash.

Economic gradient: proofs reduce entropy

- Valid proof → audit premium rebate (or insurance).
- No proof = normal audit lottery sampling.
- Outcome: "validators vote" → "validators benchmarked by reality."

agp.eth 16/24

#16

SECTION 4

Markets & Macro

Once work is measured and provable, it becomes financeable. The curve, repo, and corridor follow.

agp.eth 17/24

#17

Synthetic Labor Yield Curve

Instrument (v0): LZ(T) labor zeros

- 1 LZ(T) redeems for 1 verified α-WU credit during delivery window at maturity T.
- Price today: Z(T) (AGALPHA per α-WU credit).
- Implied yield: $r_L(T) = (1/T) \ln(50/Z(T))$, where 50 is PoC-only spot.

Why it matters

- Prices future cognitive output like energy forwards (MWRs).
- Reliability becomes a priced credit spread via tiering.
- Enables financing: sell long-dated credits to fund compute.

Launch tenors (Tier A first)

- 7D · 14D · 30D · 90D · 180D · 365D
- Settlement uses PoC-only spot index and redemption windows to prevent gaming.
- Curve prints anchored alongside GSLI prints.

agp.eth 18/24

#18

Unified Macro Stack (24-slide source)

Verbatim source slides (compressed).

Labor-Backed Compute Notes

Finance expansion by selling long-dated verified output

- Sell LZ(180D)/LZ(365D) to raise capital now.
- Deliver PoC-verified α-WU over time; redeem credits or cash.

Risk controls (make default dominated)

- Capacity-based issuance: $\text{quantile}(\alpha\text{-WU/day}) \times \beta$ safety factor.
- Performance bond + penalty buy-in on shortfall.
- Only external demand counts toward delivery (anti-wash).

Node posts bond + capacity

Mint LZ(T)

Auction raise now

Capex GPUs/runtime

Deliver PoC α-WU

Redeem credits/cash

agi.eth

18/24

#19

Labor Repo Market

Make long-dated credits liquid collateral

- Always on pool facility (borrow/lend vs LZ_A,T collateral).
- Term repo auctions (0/N, 7D, 30D) for benchmark fixings.
- Repo rates become the short end of the curve.

Conservative oracle + haircuts

- $P_{\text{oracle}} = \min(\text{median}(\text{auction}, \text{TWAP}, \text{curve}), \text{auction})$.
- Haircuts widen with volatility, illiquidity, concentration.
- Auto-convert nearing maturity into spot credits.

Suggested starting parameters (Tier A)

Collateral	cf	lt	liq bonus
LZ_A_365	0.55	0.62	6%
LZ_A_180	0.62	0.70	5%

cf = collateral factor · lt = liquidation threshold

agi.eth

20/24

#20

Standing Facility (Labor Open-Market Ops)

Corridor policy: cap stress, absorb excess liquidity

- Ceiling: lend AGIALPHA vs SR/SRCP at penalty rate.
- Floor: take AGIALPHA via reverse repo (escrow-collateralized).
- Optional: bounded DMD buys/sells to shape the long end.

Rule-bound policy (illustrative)

- Set r^* from observables: labor inflation π_L , output growth g_Q , stress d .
- Corridor widens automatically when α rises (avoid dominance).
- Hard caps; emergency actions only widen haircuts / pause new credit.

Corridor (example)



agi.eth

21/24

#21

Universal Job Paymaster → Invoices → SR/SRCP

One signature → job funded → PoC-gated payout → repay later

- Each job becomes a receivable (invoice) only after PoC finality.
- No PoC → no payout → no receivable.

Underwriting (v0)

- Start overcollateralized (AGIALPHA and/or LZ_A,T).
- Scores improve only from external-demand jobs (anti-wash).
- Pricing: $r_{\text{Borrow}} = r_{\text{Base}(t)} + s_{\text{Risk}(\text{borrower})} + s_{\text{Job}(\text{class})}$.

1) Borrower signs auth

2) Paymaster funds escrow

3) Worker executes

4) Validator attests

5) Worker paid

6) Borrower repays

From invoices to money-market collateral

- Pool PoC-final invoices (short WAM/WAL, concentration caps) → mint SR (senior) + JR (first-loss) → wrap SR into SRCP_7D / SRCP_30D for clean repo collateral.

agi.eth

22/24

#22

Publicly Verifiable Anchors

Ethereum contracts (Etherscan)

- AGIJobManager: 0x0178B6a0D66aaf908f72135B8c32fC1D5bA477
- AGIALPHA (transitory, 6d): 0x2e8f054c3e641f55f661f083c081a609eaa4e6e
- AGIALPHAEqualMinterVault: 0x27d6fE8668c6f652Ac26f8f8020D949f03aF80D8
- AGIALPHA v2 (final, 18d): 0xA61a383a130a9c20768EE8f97E21515A6046a1fA

ENS roles

- Validators: α .alpha.club.agi.eth | α .club.agi.eth
- Agents: α .alpha.agent.agi.eth | α .agent.agi.eth
- Nodes: α .alpha.node.agi.eth | α .node.agi.eth
- Sovereigns: α .alpha.agi.eth | α .agi.eth

Standards / rails (official)

- EIP-712 typed signatures (attestations + auth).
- EIP-3668 (CCIP-Read) + ENSIP-10 wildcards (namespace scale).
- ENSIP-15 normalization (anti-spoofing).
- deBridge rails for SOL ETH onboarding.

Tip: always verify domain + contract + decimals before signing.

agi.eth

23/24

#23

Build the Cathedral of Verifiable Work

Autonomy measured · Work proven · Value settled

Simple path: 1) Prove work (PoC) 2) Print truth (GSL) 3) Finance growth (curve → repo → corridor)

Repos: AGIJobsV0 · AGI-Alpha-Agent-v0 · AGI-Alpha-Node-v0

Dare to explore.

#24